(FP1) 1. Roots of polynomial equations

NO. 1

The cubic equation $x^3 + bx^2 + cx + d = 0$, where $d \neq 0$, has roots α, β, γ such that $\gamma = \frac{1}{\beta}$.

a. Show that $\beta + \frac{1}{\beta} = d - b$. [3]

b. Show also that $\beta + \frac{1}{\beta} = \frac{1-c}{d}$. [2]

c. It is given that $b = 3$, $c = -3$ and $d > 0$.

i. Find the value of d. [2]

ii. Find the value of $\alpha^2 + \beta^2 + \gamma^2$. [2]

A level FURTHER MATHEMATICS 2025/Winter/Paper1_4/Q2

Mark scheme

Question	Answer	Marks	Guidance
2(a)	$\alpha = -d$	**B1**	SOI
	$\alpha + \beta + \frac{1}{\beta} = -b$	**M1**	Uses $\alpha + \beta + \gamma = -b$ and replaces γ with $\frac{1}{\beta}$. OE
	$\beta + \frac{1}{\beta} = d - b$	**A1**	AG.
		3	

Question	Answer	Marks	Guidance
2(b)	$-d\beta+1-\frac{d}{\beta}=c$	**M1**	Uses $\alpha\beta+\beta\gamma+\gamma\alpha=c$.
	$\beta+\frac{1}{\beta}=\frac{1-c}{d}$	**A1**	AG.
		2	
2(c)(i)	$d-3=\frac{1+3}{d}\Rightarrow d^2-3d-4=0$	**M1**	Equates expressions given in **(a)** and **(b)**, with correct use of coefficients.
	$d=4$	**A1**	$d=4$ only for final answer.
		2	
2(c)(ii)	$\alpha^2+\beta^2+\gamma^2=b^2-2c$	**M1**	$\alpha^2+\beta^2+\gamma^2=(\alpha+\beta+\gamma)^2-2(\alpha\beta+\beta\gamma+\gamma\alpha)$ with correct use of coefficients.
	15	**A1**	
		2	

NO. 2

The quartic equation $x^4 + x^3 + x^2 + x + 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$.

a. Show that a quartic equation with roots $2\alpha + 1, 2\beta + 1, 2\gamma + 1, 2\delta + 1$ is $y^4 - 2y^3 + 4y^2 + 2y + 11 = 0$. [4]

The sum $(2\alpha + 1)^n + (2\beta + 1)^n + (2\gamma + 1)^n + (2\delta + 1)^n$ is denoted by S_n.

b. Find the value of S_2. [2]

c. Given that $S_3 = -22$, find the value of S_4. [2]

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Mark scheme

Question	Answer	Marks	Guidance
4(a)	$y=2x+1\Rightarrow x=\frac{1}{2}(y-1)$	**B1**	Correct substitution used in equation.
	$\frac{1}{16}(y-1)^4+\frac{1}{8}(y-1)^3+\frac{1}{4}(y-1)^2+\frac{1}{2}(y-1)+1=0$ $\frac{1}{16}(y^4-4y^3+6y^2-4y+1)+\frac{1}{8}(y^3-3y^2+3y-1)+\frac{1}{4}(y^2-2y+1)+\frac{1}{2}(y-1)+1=0$	**M1 A1**	Expands.
	$\frac{1}{16}y^4-\frac{1}{8}y^3+\frac{1}{4}y^2+\frac{1}{8}y+\frac{11}{16}=0\Rightarrow y^4-2y^3+4y^2+2y+11=0$	**A1**	Simplifies, AG.
		4	

Question	Answer	Marks	Guidance
4(b)	$S_2=2^2-2(4)$	**M1**	Uses $(\alpha+\beta+\gamma+\delta)^2=\alpha^2+\beta^2+\gamma^2+\delta^2$ $+2(\alpha\beta+\alpha\gamma+\alpha\delta+\beta\gamma+\beta\delta+\gamma\delta)$
	$S_2=-4$	**A1**	
		2	
4(c)	$S_4-2S_3+4S_2+2S_1+44=0$	**M1**	Sums over roots.
	$S_4=-76$	**A1**	
		2	

NO. 3

The cubic equation $x^3 + 2x + 1 = 0$ has roots α, β, γ.

a. Find a cubic equation whose roots are $\alpha^3 - 1, \beta^3 - 1, \gamma^3 - 1$. [3]

b. Find the value of $(\alpha^3 - 1)^2, (\beta^3 - 1)^2, (\gamma^3 - 1)^2$. [2]

c. Find the value of $(\alpha^3 - 1)^3, (\beta^3 - 1)^3, (\gamma^3 - 1)^3$. [2]

Mark scheme

Question	Answer	Marks	Guidance
a	$y = x^3 - 1$	B1	Uses correct substitution.
	$(y+1)^{\frac{3}{3}} + 2(y+1)^{\frac{1}{3}} + 1 = 0 \Rightarrow 2(y+1)^{\frac{1}{3}} = -y - 2$ $8(y+1) = -(y+2)^3 \Rightarrow 8y + 8 = -y^3 - 6y^2 - 12y - 8$	M1	Substitutes and obtains an equation not involving radicals.
	$y^3 + 6y^2 + 20y + 16 = 0$	A1	
		3	
b	$(\alpha^3 - 1)^2 + (\beta^3 - 1)^2 + (\gamma^3 - 1)^2 = 6^2 - 2(20)$	M1	Uses $\alpha^3+\beta^3+\gamma^3 = (\alpha+\beta+\gamma)^2 - 2(\alpha\beta+\beta\gamma+\gamma\alpha)$.
	−4	A1	FT
		2	
c	$(\alpha^3 - 1)^3 + (\beta^3 - 1)^3 + (\gamma^3 - 1)^3 = -6(-4) - 20(-6) - 16(3)$	M1	Uses their equation from (a) or a valid formula for the sum of cubes.
	96	A1	
		2	

Comment

a. This was mostly done well. There were two main methods used. Substitution of a surd followed by simplification, or rearrangement to a form including x^3 followed by substitution, were equally successful. Most demonstrated understanding of the need to isolate the surd term prior to cubing.

A few tried using the roots of the original equations to create the roots of the new equation but were largely unsuccessful.

b. The result usually followed correctly from their answer for **part (a)**.

c. Most candidates used $S_3 + 6S_2 + 20S_1 + 16S_0 = 0$ accurately.

A few tried to use formulae but these formulae were often wrongly remembered, or incorrect values were used.

NO. 4

It is given that

$$\alpha + \beta + \gamma + \delta = 2,$$
$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 3,$$
$$\alpha^3 + \beta^3 + \gamma^3 + \delta^3 = 4,$$

a. Find the value of $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta$. [2]

b. Find the value of $\alpha^2\beta + \alpha^2\gamma + \alpha^2\delta + \beta^2\alpha + \beta^2\gamma + \beta^2\delta + \gamma^2\alpha + \gamma^2\beta + \gamma^2\delta + \delta^2\alpha + \delta^2\beta + \delta^2\gamma$. [3]

c. It is given that α, β, γ, δ are the roots of the equation

$$6x^4 - 12x^3 + 3x^2 + 2x + 6 = 0.$$

i. Find the value of $\alpha^4 + \beta^4 + \gamma^4 + \delta^4$. [3]

ii. Find the value of $\alpha^5 + \beta^5 + \gamma^5 + \delta^5$. [2]

Mark scheme

Question	Answer	Marks	Guidance
a	$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = (\alpha + \beta + \gamma + \delta)^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$ $3 = 2^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$	M1	Substitutes into formula for sum of squares.
	$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = \frac{1}{2}$	A1	
		2	
b	$\alpha^2(\beta + \gamma + \delta) + \beta^2(\alpha + \gamma + \delta) + \gamma^2(\alpha + \beta + \delta) + \delta^2(\alpha + \beta + \gamma)$ $= \alpha^2(2 - \alpha) + \beta^2(2 - \beta) + \gamma^2(2 - \gamma) + \delta^2(2 - \delta)$	M1 A1	Factorises and substitutes.
	$2(3) - 4 = 2$	A1	
		3	
c.i	$6S_4 - 12S_3 + 3S_2 + 2S_1 + 24 = 0 \Rightarrow 6S_4 - 12(4) + 3(3) + 2(2) + 24 = 0$	M1	Sums and substitutes.
		A1	
	$S_4 = \frac{11}{6}$	A1	
		3	
c.ii	$6S_5 - 12S_4 + 3S_3 + 2S_2 + 6S_1 = 0 \Rightarrow$ $6S_5 - 12\left(\frac{11}{6}\right) + 3(4) + 2(3) + 12 = 0$	M1	Multiplies equation through by x, sums and substitutes. $S_5 - 2S_4 + \frac{1}{2}S_3 + \frac{1}{3}S_2 + S_1 = 0$ $\Rightarrow S_5 - 2\left(\frac{11}{6}\right) + \frac{1}{2}(4) + \frac{1}{3}(3) + 2 = 0$
	$S_5 = -\frac{4}{3}$	A1	
		2	

Comment

a. Almost all candidates recalled and applied the correct formula.

b. The simplest solution was to recognise that $\beta + \gamma + \delta = 2 - \alpha$ and so the first three terms can be simplified to $2\alpha^2 - \alpha^3$, for example. Other successful solutions used known formulas. Those candidates who tried to evaluate $\sum\alpha\sum\alpha^2$ often made algebraic mistakes.

c. i. The majority found this straightforward. The summation of the equations was done extremely well, the only common problem was to forget that there were 4 roots.

ii. A correct answer in **(i)** was usually followed by a correct solution here.

NO. 5

The cubic equation $2x^3 + x^2 - px - 5 = 0$, where p is a positive constant, has roots α, β, γ.

a. State, in terms of p, the value of $\alpha\beta + \beta\gamma + \gamma\alpha$. [1]

b. Find the value of $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$. [2]

c. Deduce a cubic equation whose roots are $\alpha\beta$, $\beta\gamma$, $\alpha\gamma$. [1]

d. Given that $\alpha^2 + \beta^2 + \gamma^2 = \frac{1}{3}$, find the value of p. [2]

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Mark scheme

Question	Answer	Marks	Guidance
a	$-\frac{1}{2}p$	B1	
		1	
b	$\alpha\beta^2\gamma + \alpha^2\beta\gamma + \alpha\beta\gamma^2 = \alpha\beta\gamma(\beta+\gamma+\alpha) = \frac{5}{2}\left(-\frac{1}{2}\right)$	M1	Factorises.
	$-\frac{5}{4}$	A1	
		2	
c	$4z^3 + 2pz^2 - 5z - 25 = 0$	B1 FT	OE FT on *their* value of $\alpha\beta + \beta\gamma + \gamma\alpha$ (in terms of p).
		1	
d	$\frac{1}{3} = \frac{1}{4} + p$	M1	Uses $\alpha^2 + \beta^2 + \gamma^2 = (\alpha+\beta+\gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$
	$p = \frac{1}{12}$	A1	CAO
		2	

Comment

Candidates were able to employ standard methods of dealing with the roots of equations well.

a. This was almost always correct.
b. Most candidates factorised the expression correctly, however some made sign errors when substituting so did not achieve full credit.
c. A minority of candidates realised that the previous two question parts had given them two of the coefficients needed and could write down the required equation immediately. Some candidates attempted to use a substitution of $w = z^2$ however this was unsuccessful. Many candidates did not attempt the question.

d. Most candidates were able to form and solve the equation for p correctly.